

Detour: Computing Limb Lengths in the Neighbor-Joining Algorithm | Step 1

In the main text, we were attempting to assign limb lengths to leaves of a tree constructed from an arbitrary distance matrix. We set the limb length of i equal to $1/2(D_{ij} + \Delta_{ij})$ and the limb length of j equal to $1/2(D_{ij} - \Delta_{ij})$, where

$$\Delta = \frac{TotalDistance_D(i) - TotalDistance_D(j)}{n - 2}$$

Where do these formulas come from? Assume for a moment that D is an additive matrix, and select some leaf k that is not equal to i or j . If m is the parent of i and j , then we already have the limb length formula

$$LimbLength(i) = (1/2) (D_{i,k} + D_{m,k} - D_{j,k})$$

As a result, it might seem like we should use this formula in the neighbor-joining algorithm for an arbitrary distance matrix D . However, if D is non-additive, then the expression $(D_{ij} + D_{i,k} - D_{j,k})/2$ will vary depending on how we select k . So we need a formula that still computes $LimbLength(i)$ when D is additive but that also gives us a single value when D is non-additive. To this end, we can compute the average of the above formula over all choices of $n - 2$ leaves k :

$$\frac{1}{n - 2} \sum_{\text{all leaves } k \text{ differing from } i \text{ and } j} \frac{D_{i,j} + D_{i,k} - D_{j,k}}{2}$$